A degree-316 Diophantine consistency equation in twelve positive unknowns

Research supplement for ProveIt

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1 A degree-316 ordinary-strong merger

Result and scope. There is an explicit equation in twelve strictly positive integer unknowns, of exact total degree 316, evaluated in 102 binary arithmetic operations (61 multiplications and 41 additions or subtractions), whose positive unsolvability is equivalent in ZFC to Con(ZFC). The three-equation system has residual degrees $2, 26, 288$ in 95 operations, or 94 after the inherited positive transport-coordinate shift. A sum of squares has degree 576 in 103 operations. Fixed numeral leaves and reuse are free; multiplication by a numeral is charged.

This improves the authenticated degree-897, 108-operation twelve-positive construction on both measures. It does not reduce the eleven-positive variable frontier, assert global optimality, prove consistency, or provide a new end-to-end proof-assistant certificate. The native compiler and literal K215/refutation calibration remain explicit inherited dependencies.

The structural change is to retain f and eliminate the ordinary strong coordinate i, together with τ and y. Thus the crucial polynomial is $Q = \Delta(f^2 - 1)$ of degree 14, not the normalized $\Delta^2(ic^2)^2$ of degree 46. A separated three-root sum recovers rationality; rational Pell saturation recovers integrality. Removing a known common denominator factor from its norm gives a degree-288 inner polynomial.

1.1 Literal polynomial

The supplied positive variables are

$$J, F, \alpha, z, f_0, T, s, w, \eta, \zeta, \gamma, t.$$

Use exactly the entire authentic source31 Wang943 121-by-8801 coefficient tuple (β, K_0, M_C, M_F), with $\beta = 2^{1064921} - 1$ and M_F already the shifted mask. These are fixed integers, not unknowns. The payloads, hashes, tuple adapter, and inherited ordinary proof are included in the packet. Every other name below is a polynomial abbreviation:

$$\begin{aligned}
q &= \beta J + 1, & C_w &= q - F - \alpha, & X &= wq, & Y &= sq^3,\\
E &= XY, & k &= \eta + \zeta, & c &= kY + \eta, & a &= E + Y,\\
H &= 4a + 3, & \Delta &= a^2 + H = (a+2)^2 - 1, & \ell &= EkY,\\
A &= 1 + \ell(\ell + k), & G &= 1 + \Delta c^2, & b_0 &= X + ac.
\end{aligned}$$

Define

$$
\begin{aligned}
R &= ((q-1)(q-F)+\alpha)(q^2-1)+(M_C+qM_F)J,\\
N_t &= (K_0+w)C_w+q-F-z(q-1),\\
U &= 4q^4, \qquad u = k-R-1,\\
D_* &= Ub_0+H(U\gamma-u), \qquad N_m = D_*^2-U^2G,\\
f &= f_0+1, \qquad Q=\Delta(f^2-1), \qquad d=Q-1,\\
V &= c(Tf-1)-Rf^2, \qquad E_a = QV^2-1, \qquad M_2 = 2f\ell,\\
Z &= c^4M_2^2E_a, \qquad x=c^2t, \qquad r=c^4f^2A,\\
M &= d(x^2+Q+r)-Z,\\
C_0 &= M^2+4d^2\{x^2(r-Q)-Qr\}, \qquad B = 2dQ-M,\\
N &= C_0^2-16d^2x^2rB^2.
\end{aligned}
$$

The equation is

$$\boxed{N_t(1+N_m)(1+N)-1=0.} \tag{1}$$

The corresponding system is $N_t=1$, $N_m=0$, $N=0$. The sum of squares is $(N_t-1)^2+N_m^2+N^2=0$.

The inner norm is exactly

$$N = \prod_{\rho^2=Q,\ \tau^2=A} [d(c^2t-\rho-c^2f\tau)^2-c^4M_2^2E_a], \tag{2}$$

with both formal signs of each root. To verify this identity, put $b=c^2f\tau$, so $b^2=r$. Each factor is $M-2dx\rho-2dxb+2d\rho b$. Its norm in ρ is $C_0+4dx(2dQ-M)b$; its norm in b is the stated N. This is a polynomial-ring identity, including degenerate specializations. It does not assume independent Galois automorphisms after specialization.

For comparison, the eight-factor linear norm of

$$c^2dt-d\rho-c^2df\tau-c^2M_2\sqrt{dE_a}$$

is d^4N. Eliminating the two signs of the last root proves this identity and explains the denominator deflation. The actual circuit has no division.

1.2 Unit guard and the main root

Since U is divisible by 4,

$$N_m \equiv D_*^2 \pmod{16}, \qquad N \equiv C_0^2 \pmod{16}.$$

At a root of (1), all three factors are integer units. Neither norm can equal -2, which is not a square modulo 4. Hence $N_m=N=0$ and $N_t=1$.

The authentic transport/mask interface then gives

$$q\ge 16, \quad C_w>0, \quad 3q<R<q^4-2, \quad X\ge q, \quad Y\ge q^3, \quad E\ge q^4>R+2,$$

$$k\ge 2, \qquad kY<c<k(Y+1), \qquad a+2<2E.$$

In particular $C_w\le 0$ would imply $N_t\le 1-F-(z-1)(q-1)\le 0$, so it is impossible. The shift $f=f_0+1$ ensures $f\ge 2$, $Q>1$, and $d>0$; no $f=1$ or zero-denominator exception is discarded.

From $N_m = 0$, the rational root $g = D_*/U$ of $g^2 = G$ is an integer. It is positive: $H < 5a$, $b_0 > akY$, and $u < k$ give

$$D_* > Ub_0 - Hk > ak(UY - 5) > 0.$$

Reduction of $D_* = Ug$ modulo U gives

$$U \mid H(k - R - 1). \tag{3}$$

Neither the omitted index quotient nor the old main quotient has yet been recovered.

1.3 Rationality from separated square classes

Write $\rho = \sqrt{Q} > 0$ and $\tau = \sqrt{A} > 0$. If $V = 0$, then $E_a = -1$ and every real factor in (2) is strictly positive. Thus $V \neq 0$. Since V is an integer, $E_a \geq Q - 1 = d$ and

$$y = \sqrt{E_a/d} \geq 1.$$

A vanishing factor gives signs $\epsilon_j \in \{1, -1\}$ with

$$t = \epsilon_1 \frac{\rho}{c^2} + \epsilon_2 f\tau + \epsilon_3 M_2 y. \tag{4}$$

Before any rationality conclusion, one has

$$\rho < (a+2)f, \qquad \tau < \ell + k, \qquad \ell - k = k(EY - 1) \geq 2EY - 2 > 2E > a + 2.$$

Consequently

$$0 < \frac{\rho}{c^2} < f\ell < f\tau, \qquad \frac{\rho}{c^2} + f\tau < 2f\ell = M_2 \leq M_2 y. \tag{5}$$

Each term in (4) is a rational multiple of a square root of a positive integer; use $y = \sqrt{dE_a}/d$. Distinct squarefree radicals are linearly independent over $\mathbb{Q}$. In a rational signed sum, every irrational square class must cancel. The largest term exceeds the sum of both others, so its class cannot cancel unless y is rational. Then the strict inequality between the two remaining terms excludes their cancellation in an irrational class. Hence ρ, τ, y are rational. In particular ρ and τ are integers. This argument explicitly allows coincident square classes.

Rational Pell saturation. For integers $m \geq 2, V$ and rational y, the equation

$$m^2V^2 - (m^2 - 1)y^2 = 1$$

forces y to be integral. Indeed, write $y = b/r$ in lowest terms. Then $r^2 \mid m^2 - 1$. For the positive nonsquare integer $D_0 = (m^2 - 1)/r^2$, the pairs (m, r) and $(m|V|, |b|)$ are integer-coordinate Pell solutions. If their fundamental-unit indices are j, l, divisibility of their abscissas forces l to be an odd multiple of j: reduce modulo $2j$, reflect into $[0, j]$, and use strict growth with $\chi_{n+2j} \equiv -\chi_n \pmod{\chi_j}$. Ordinate divisibility then gives $r \mid b$, so $r = 1$. The original equation excludes zero-coordinate cases.

Apply this lemma with $m = \rho$ to

$$\rho^2V^2 - (\rho^2 - 1)y^2 = 1.$$

Thus y is integral. Equation (4) now makes ρ/c^2 integral. Put $i = \rho/c^2 > 0$. The recovered equations are

$$\tau^2 = 1 + \ell(\ell + k), \qquad (ic^2)^2 = \Delta(f^2 - 1), \qquad QV^2 - (Q - 1)y^2 = 1.$$

All square-root signs may be made positive. No cancellation of Δ and no ordinary-rank assumption has occurred in this recovery.

1.4 Ordered index recovery

This part uses the preserved ordinary strong-rank theorem and the source50 absorber, with their hypotheses now verified. Let $A_0 = a + 2$ and $P_0 = 2EY + 1$. The two recovered Pell norms yield indices $p, n \geq 1$ with

$$g = \chi_{A_0}(p), \quad c = \psi_{A_0}(p), \quad \tau = \chi_{P_0}(n), \quad k = 2\psi_{P_0}(n).$$

The first norm's fundamental integer-coordinate solution is $(2EY+1, 2)$ for discriminant $EY(EY+1)$. Since $P_0 > A_0$ and $c > k$, monotonicity gives $p > n$.

The Pell recurrence gives $k \equiv 2n \pmod{4EY}$, and $4EY = 4ws^2q^7$ is divisible by U. Set $\mu = U/\gcd(U, H)$. The number H is odd; a prime common to q, H must be 3. If $3 \mid q$, then $H \equiv 3 \pmod{27}$, so its 3-adic valuation is exactly one. Hence

$$\gcd(U, H) \leq 3, \qquad \mu \geq \frac{4q^4}{3} > R + 1.$$

Together with (3) this gives $2n = R + 1 + j\mu$, where positivity of n forces $j \geq 0$. Thus $n \geq 25$, $p \geq 26$, and Pell growth proves

$$c > A_0\Delta^2, \qquad c > 2p, \qquad c > 2R.$$

The ordinary strong-rank theorem now applies to $(ic^2)^2 = \Delta(f^2 - 1)$. It gives an index m such that

$$f = \chi_{A_0}(m), \quad p \mid m, \quad c \mid m, \quad m > 2p, \quad ic^2 = \Delta\psi_{A_0}(m), \quad c^2 \mid f^2 - 1.$$

The theorem handles the possibly nonfundamental parameter in the underlying squarefree Pell field. Its proof and the signed auxiliary argument are included in the preserved ordinary source, Sections 4–5.

The odd auxiliary index j_0 satisfies $j_0 \equiv \pm p \pmod{m}$ from $V \equiv -c \pmod{f}$, and $j_0 \equiv \pm R \pmod{c}$ from $Q \equiv 0 \pmod{c}$, $V \equiv -R \pmod{c}$. Since $c \mid m$ and $0 < p, R < c/2$, it follows that $p = R$. The signed reduction uses $2\chi_{A_0}(m-1) < \chi_{A_0}(m)$, not an assumed sign of V.

Now $n < R$, so $|2n - R - 1| < R < \mu$ forces $2n = R + 1$. Hence $u = k - 2n$ is divisible by $4EY$, and therefore by both E, U. Set

$$h = u/E, \qquad \gamma_{\text{old}} = \gamma - u/U.$$

The first is positive because $n \geq 25$ and $\psi_{P_0}(n) > n$. The main carrier gives $\gamma_{\text{old}} = (g - b_0)/H$. Since $n \geq 2$, $k \geq 4P_0 > X$, so $c > X$ and $g - b_0 > (a+1)c - ac - X = c - X > 0$. All positive ordinary compiler coordinates are restored. Its inherited soundness theorem extracts a blank K215 halt and a finite ZFC refutation.

1.5 Completeness

Conversely take a canonical positive ordinary compiler witness. Its index identity is $p = R = 2n - 1$, so u is a positive multiple of $4EY$ and of U. Set

$$\gamma = \gamma_{\text{old}} + u/U, \qquad f_0 = f - 1, \qquad t = i + f\tau + M_2 y.$$

These are positive integers. Then $D_* = Ug$, while the positive-sign factor in (2) is $c^4M_2^2(dy^2 - E_a) = 0$. Thus $N_m = N = 0$ and $N_t = 1$. The preserved change $i_{\text{ordinary}} = \Delta i_{\text{normalized}}$ provides ordinary witnesses from normalized ones when necessary. This proves existence equivalence, hence the claimed consistency equivalence.

The shifted system uses $(K_0 + w)C_w = F + z(q-1)$, corresponding to $z_{\text{old}} = z + 1$. Canonical witnesses have $X = 2^R$ and $w > q > F$, which force $z_{\text{old}} \geq 2$. Thus the new z is positive, and the shift saves one operation.

1.6 Exact degree and replay

Put

$$q_0 = \beta J, \quad Y_0 = sq_0^3, \quad E_0 = wq_0Y_0, \quad c_0 = (\eta + \zeta)Y_0.$$

These have degrees $1, 4, 6, 5$. The inner leading forms are

$$Q_{14} = E_0^2 f_0^2, \quad V_7 = c_0 T f_0, \quad (M_2)_{12} = 2f_0 E_0 c_0, \quad Z_{72} = 4E_0^4 c_0^8 T^2 f_0^6.$$

All other terms of M have degree at most 58. Therefore

$$N_{288} = Z_{72}^4 = 2^8 E_0^{16} c_0^{32} T^8 f_0^{24}.$$

The subtracted term in N has degree at most 238, so there is no possible cancellation of this highest form. For the main residual use

$$N_m = U^2\{(b_0 + H\gamma)^2 - G\} - 2UHu(b_0 + H\gamma) + H^2u^2.$$

The bracket has degree 18 and highest form $8E_0^2 c_0\gamma$; the other terms have degree at most $25, 20$. Thus

$$(N_m)_{26} = 128q_0^8 E_0^2 c_0\gamma.$$

Since $(N_t)_2 = w(q_0 - F - \alpha) - zq_0$ is nonzero,

$$P_{316} = 2^{15} q_0^8 E_0^{18} c_0^{33} \gamma T^8 f_0^{24}[w(q_0 - F - \alpha) - zq_0]$$

is nonzero for every authentic $\beta > 0$. These are exact fixed-instance degrees, not generic numerical tests.

The packet's independent arithmetic checker verifies 36 exact signed norm/deflation identities, 100 modular identities, both residue guards, 144 pre-index bounds, and 24 complete finite-field polynomial rays including the actual fixed coefficients. It also verifies payload hashes, analytic leaders, and transport shifts. Both ordinary Python and `python -O` runs pass. The inert circuits have no dead gates or inputs. Finite checks validate identities, counts and degrees, not the inherited unbounded compiler theorem. The work is AI-assisted and makes no claim of independent human review or kernel-certified consistency reduction.